

UD65640

Application Note

OTP

AN-K0009-260107

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1. Target

Introduce OTP register and OTP flow.

2. Register

OTP Ctrl: OTP Control (82h)

82h	OTP Ctrl									
	W / R	B7	B6	B5	B4	B3	B2	B1	B0	Hex
Command	—	1	0	0	0	0	0	1	0	82h
parameter 1	W / R	GRP8	GRP7	GRP6	GRP5	GRP4	GRP3	GRP2	GRP1	00h
parameter 2	W / R	0	GRP15	GRP14	GRP13	GRP12	GRP11	GRP10	GRP9	00h
parameter 3	W / R	0	0	0	0	GRP19	GRP18	GRP17	GRP16	00h
parameter 4	W / R	0	0	0	OTPEN	0	OTP_ACT_MODE[2:0]			00h
Description	OTP_ACT_MODE									
	Operation mode for OTP. UD65640 only supports program OTP in sleep-in mode.									
	OTP_ACT_MODE[2:0]		Function							
	3'b000		Program group							
Description										
	OTPEN									
	Enable for OTP access, please be sure to set OTP control registers and data before enable OTP access.									
	1: Enabled.									
	0: Disabled.									
Description										
	GRP1 ~ GRP19: Enable requires program the group.									
	1: Enabled.									
	0: Disabled.									
Description										
	1 = Must fill 1; 0 = Must fill 0; X = Don't care.									
Restriction	Note:									

OTP State: OTP State (83h)

84h	OTP State									
	RO	B7	B6	B5	B4	B3	B2	B1	B0	Hex
Command	—	1	0	0	0	0	0	1	1	83h
parameter 1	RO	0								00h
parameter 2	RO	0	0	G19_WC	G15_WC	WRALLFIN	0	0	LOADALLFIN	00h
parameter 3	RO	G4_WC[2:0]			G3_WC[1:0]		G2_WC[1:0]		G1_WC	00h
parameter 4	RO	G14_WC	G7_WC[2:0]			G6_WC[1:0]		G5_WC[1:0]		00h
parameter 5	RO	G13_WC	G12_WC[1:0]		G11_WC	G9_WC[1:0]		G8_WC[1:0]		00h
parameter 6	RO	0	0	G10_WC[5:0]						00h
Description	LOADALLFIN Load-all-data from OTP finish indicator. It will clear after read. WRALLFIN Write-all operation finish indicator. It will clear after read. G1_WC[1:0] Group 1 read the number of burn-ins, max is 1. G2_WC[1:0] Group 2 read the number of burn-ins, max is 2. G3_WC[1:0] Group 3 read the number of burn-ins, max is 2. G4_WC[1:0] Group 4 read the number of burn-ins, max is 3. G5_WC[1:0] Group 5 read the number of burn-ins, max is 2. G6_WC[5:0] Group 6 read the number of burn-ins, max is 2. G7_WC[2:0] Group 7 read the number of burn-ins, max is 3. G10_WC[5:0] Group 10 read the number of burn-ins, max is 6. G11_WC Group 11 read the number of burn-ins, max is 1. G12_WC[1:0] Group 12 read the number of burn-ins, max is 2. G13_WC Group 13 read the number of burn-ins, max is 1. 1 = Must fill 1; 0 = Must fill 0; X = Don't care.									

OTP Power: OTP Power (85h)

85h	OTP Power									
	W / R	B7	B6	B5	B4	B3	B2	B1	B0	Hex
Command	—	1	0	0	0	0	1	0	1	85h
parameter 1	W / R	0	0	0	0	0	0	REF_OTP_EN	VREFP_OTP_EN	00h
Description	REF_OTP_EN: Enable OTP reference voltage before program OTP. 1: Enabled. 0: Disabled. VREFP_OTP_EN: 1: Enabled OTP VREFP voltage before program OTP. 0: Disabled OTP VREFP voltage before program OTP. Note: UD65640 only supports program OTP in sleep-in mode, so REF_OTP_EN and VREFP_OTP_EN needs to be started in sleep mode. 1 = Must fill 1; 0 = Must fill 0; X = Don't care.									
Restriction										

OTP State 1: OTP State 1 (87h)

87h	OTP State 1									
	RO	B7	B6	B5	B4	B3	B2	B1	B0	Hex
Command	—	1	0	0	0	0	1	1	1	87h
parameter 1	RO	0	0	G18_WC[1:0]		G17_WC[1:0]		G16_WC[1:0]		00h
Description	G16_WC[1:0] Group 16 read the number of burn-ins, max is 2. G17_WC[1:0] Group 17 read the number of burn-ins, max is 2. G18_WC[1:0] Group 18 read the number of burn-ins, max is 2.									
Restriction										

3. OTP Program Flow(Version UD65640B):

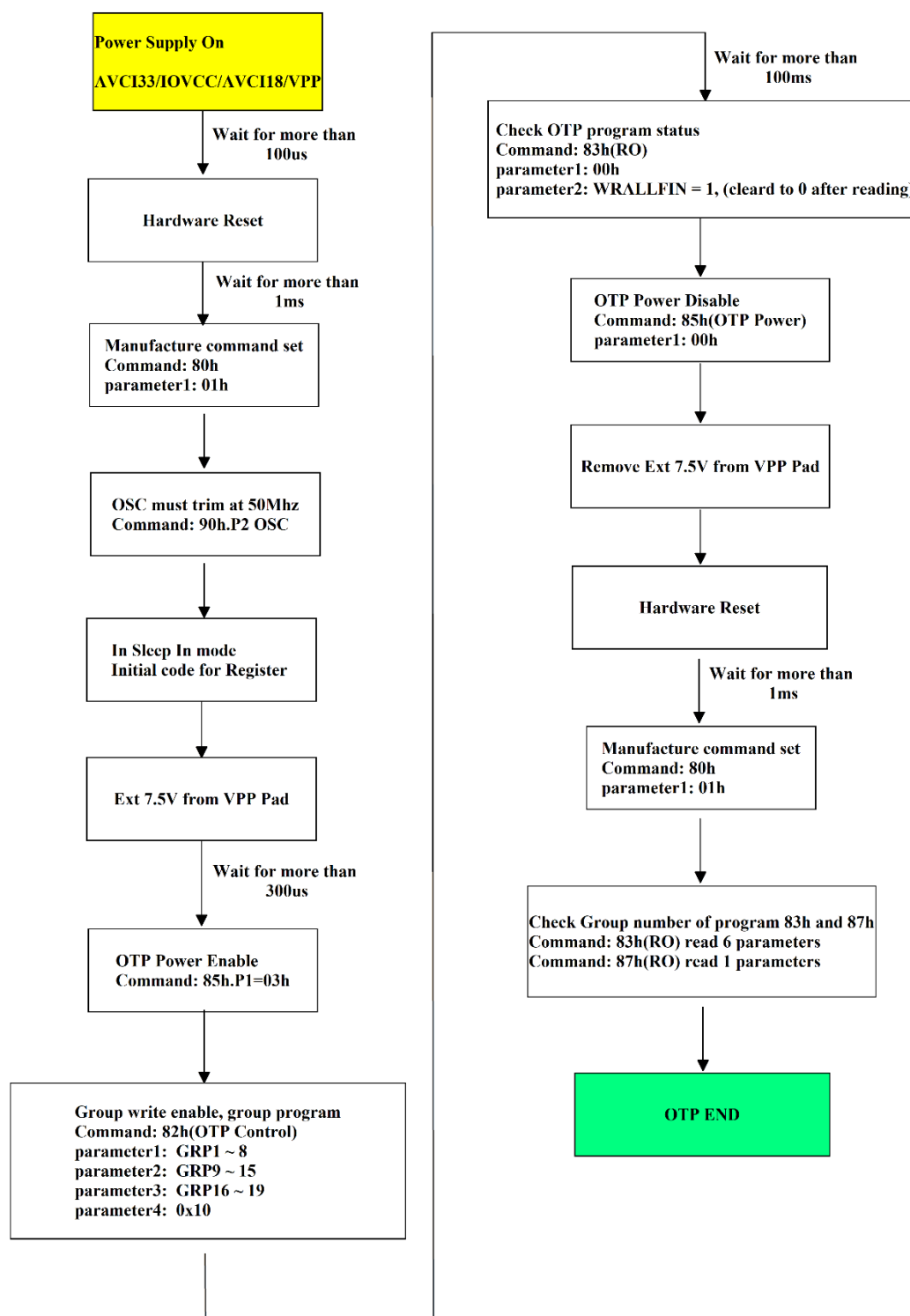


Figure 1 – OTP Flow

4. Gamma OTP Program:

Gamma point at OTP Group 10, Can be burned 6 times in total.

Gamma mapping at OTP Group 7, Can be burned 3 times in total.

4.1 Gamma Group 7 and Group 10 Link

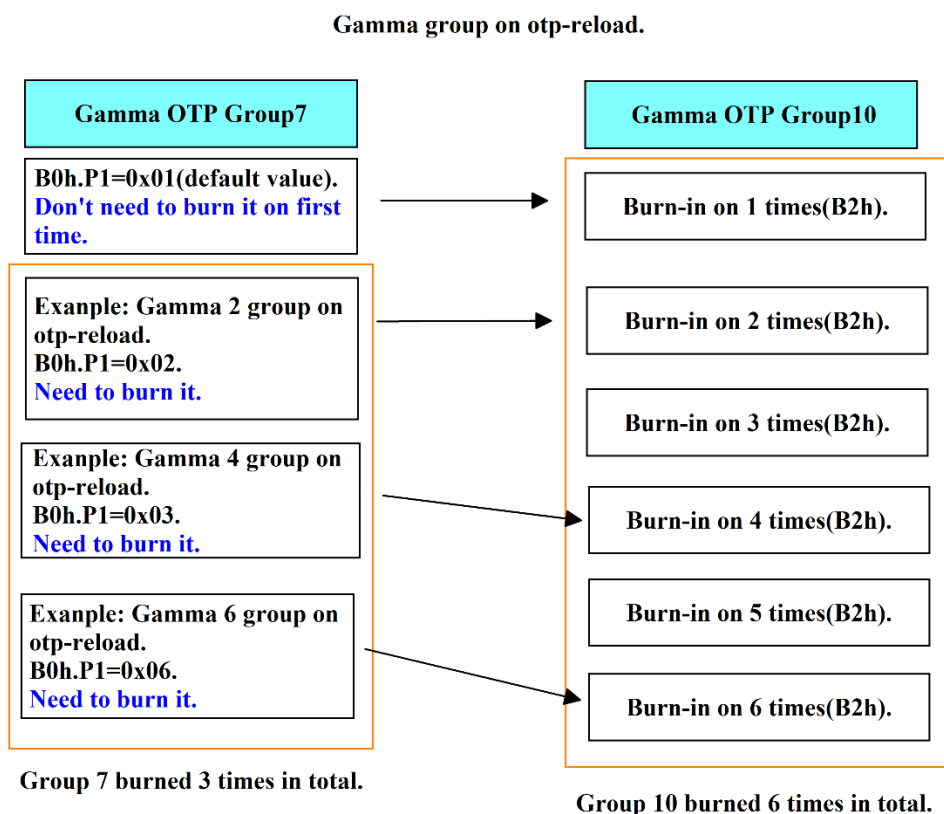


Figure 2 – Gamma Group 7 and Group 10 link

4.2 OTP Gamma Mapping Group 7 Program

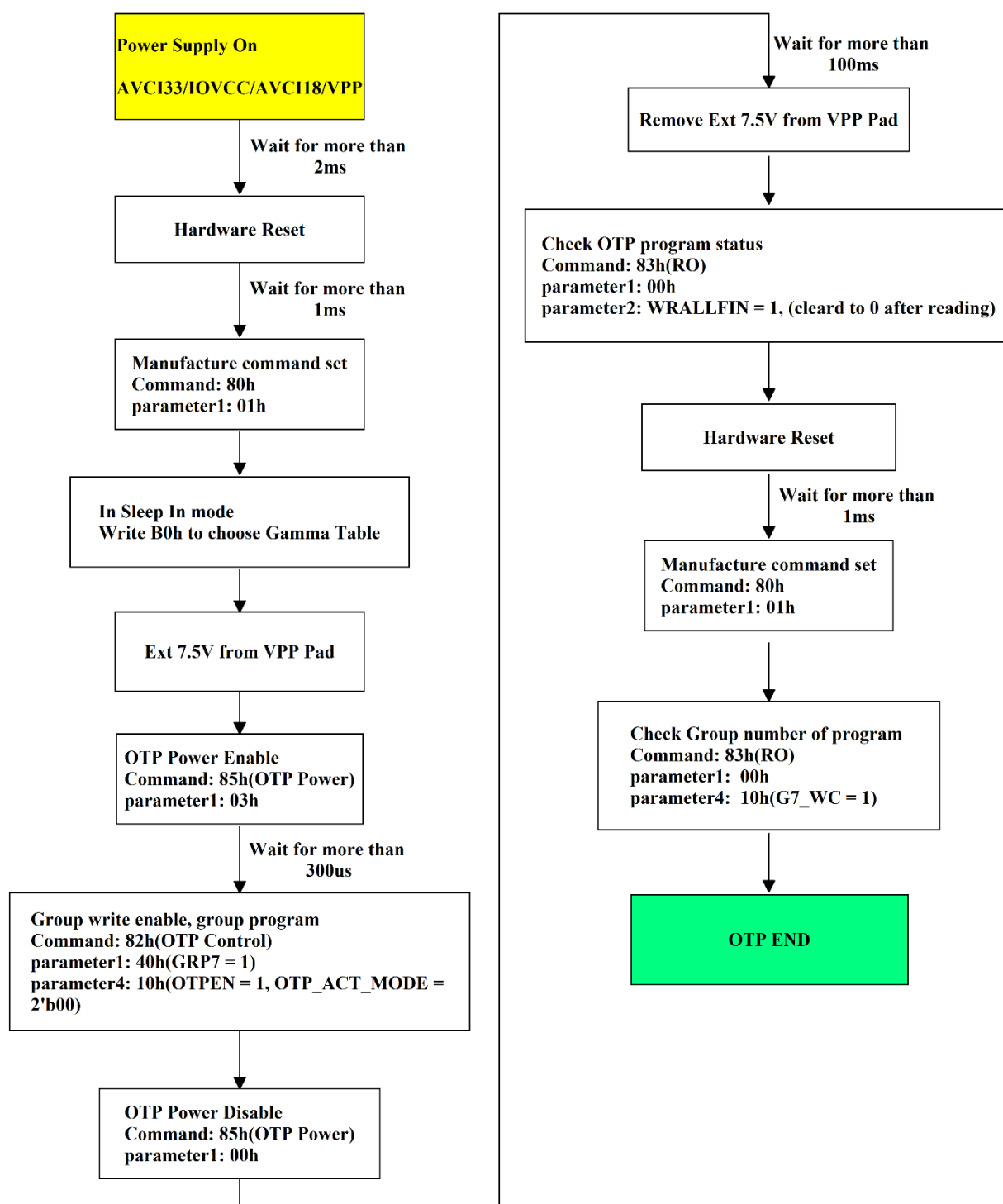


Figure 3 – OTP Group 7 Program

4.3 OTP Gamma Point Group 10 Program

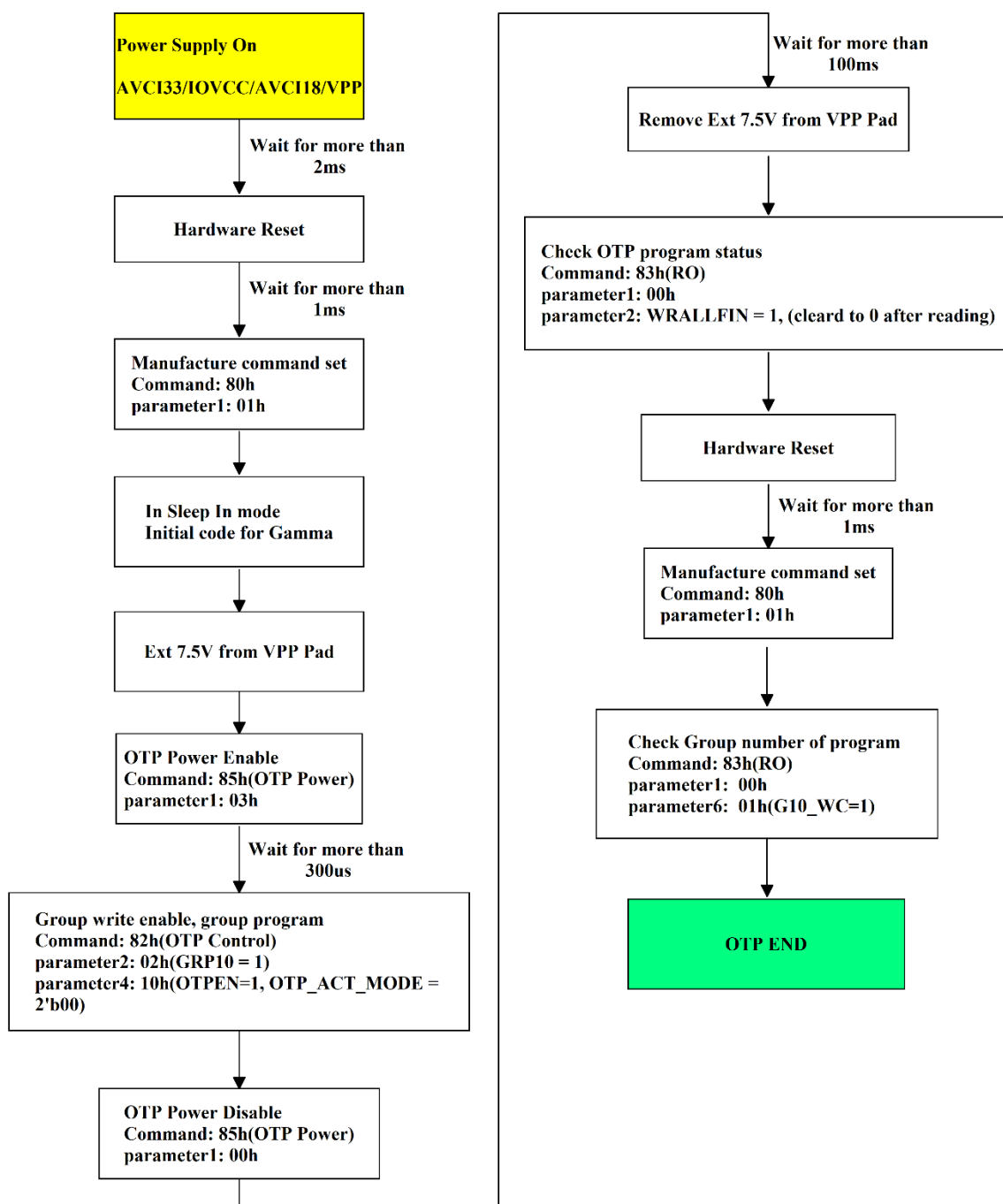


Figure 4 - OTP Group 10 Program

5. OTP Group Table

Command Name	Code	OTP Group	Max numbers of program
Device Code	88h	Group 1	1
Interface Control	8Dh	Group 1	
AVEE Control	96h	Group 4	3
VEMR Control	97h	Group 4	
SWIRE Control	98h	Group 5	2
Image Engine Control	A0h	Group 6	2
Panel Driving Settings1	A1h	Group 6	
Panel Driving Settings2	A2h	Group 6	
Digital Gamma Table Settings index	B0h	Group 7	3
Digital Gamma Setting values	B2h	Group 10	6
TS Control	C0h	Group 9	2
TS Trim	C2h	Group 9	
TC Settings	C3h	Group 9	
TP Settings	C6h	Group 9	
TC LUT	C5h	Group 12	2

6. How to confirm the number of OTP burning times

Example Gamma Group 10.

Register: 83h
Read G10_WC[5:0] = 0x01 (Burn In 1 times)
Read G10_WC[5:0] = 0x02 (Burn In 2 times)
Read G10_WC[5:0] = 0x04 (Burn In 3 times)
Read G10_WC[5:0] = 0x08 (Burn In 4 times)
Read G10_WC[5:0] = 0x10 (Burn In 5 times)
Read G10_WC[5:0] = 0x20 (Burn In 6 times)

7. Revision History

Version No.	Date	Page	Description
AN-K0009-240815	2024/8/15	10	OTP Application Note
AN-K0009-240912	2024/9/12	10	Modify OTP program flow.
AN-K0009-250507	2025/5/07	10	Modify OTP program flow(Version UD65640B).
AN-K0009-260107	2026/1/07	10	Add TS, TC, TP Group